

AI 504: Knowledge representation

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Spring 2026 syllabus

Course overview

It is an overgeneralization—but a useful one—to say that models of artificial intelligence come in either a *neural* or *symbolic* flavor, depending on what type of biological cognition they are trying to mimic. Neural or “deep learning” approaches mimic rapid and unconscious cognitive processes like perception and sensation. Symbolic or “logical” approaches mimic slower and deliberate cognitive processes like reasoning and calculation. Over the brief history of AI as a field, these two have established themselves as the dominant paradigms of machine intelligence, and any serious students of the field (such as yourselves!) must become acquainted with both.

This course is an introduction to symbolic methods. These constitute a vast body of knowledge that cannot begin to be squeezed into a single semester. Therefore our focus is on teaching *principles* rather than any attempt at a comprehensive survey. In particular, we focus on the problem of *inference*, which is, succinctly:

Given that we know X, when can we conclude that Y must hold too?

To a logician or philosopher, this would be known as the problem of *consequence* or *entailment*, but it all comes to the same thing. Like many AI tasks, the question of inference is more generally applicable than this strict phrasing might suggest. (For example, it is perhaps technically correct but overly reductive to say that LLMs “only” solve the problem of next-bit prediction.)

There are, again, two broad approaches to solving the problem of inference:

- *Model-theoretic*, i.e., “we cannot imagine a world in which X is true but Y is not,” and
- *Proof-theoretic*, i.e., “we can construct a chain of reasoning starting at X and ending at Y .”

Of course this being *artificial* intelligence, we not only have to say (in a precise and mathematical sense) what we mean by “worlds,” “chains of reasoning,” etc., we actually have to be able to *implement* these on a computer. That is exactly what we do in this course, starting in the context of some very simple logical systems, and building to the more sophisticated first-order logic. Along the way we will actually implement solutions to the problem of inference in many of these logics on the computer. We will observe that more expressive logics generally require more complex solutions, past a point becoming literally undecidable.

So if I were forced to give a pithy summary of the main themes in a few bullets, here’s how it would go:

- Represent proofs and models as structured data. (This is literally the *knowledge representation*.)
- Relate model-theoretic and proof-theoretic consequence.
- Relate expressive power and the complexity of inference.
- Implement solutions to the problem of inference when possible.

There is a whole universe of knowledge representation-related buzzwords that we will not touch. (For example: *ontology*, *logic programming*, *forward and backward chaining*, *description logics*, etc.) What this course gives you are the *tools to teach these things to yourself*, by giving a thorough grounding in the foundations of the subject. While this course relies on nothing beyond elementary discrete mathematics, it demands a certain mathematical maturity that you will develop throughout the semester and that will serve you well for the rest of your academic and professional life. Come ready to think deeply about simple things!

Course mechanics

Instructor. Siddharth Bhaskar (that's me); my email address is `bhaskar@imada.sdu.dk`. You are welcome to email me, but most technical questions should be reserved for office hours (see below). I do not check or respond to messages on itsLearning.

Teaching assistants. Marco Santamaria (initially); Stefano Volpe (subsequently)

Study group. You are expected to have a network of acquaintances in the class that you can work together with and rely on in case of a missed class. I understand this is standard practice here, but please see me if you have concerns.

Textbook and course website. Our textbook is *Logic from Language* by Larry Moss. It is still in the form of a draft manuscript. We will use the university-provided itsLearning website to store this syllabus, the textbook, and assign and submit problem sets.

Language. I will speak and write in English to you, but you are welcome to communicate in either English or Danish to me. For example, I do not anticipate having much trouble reading problem sets written in Danish. If you have trouble understanding me, please don't hesitate to stop and say so!

In-person mechanics

There are three modes of face-to-face interaction in this course: *lecture*, *recitation*, and *office hours*. (*Lecture* and *recitation* correspond respectively to *holdtime* and *fællestime*.)

Lecture. There is one 90-minute lecture each week. Lectures typically consist of me lecturing at the chalkboard, but this pattern might see some variation. In lecture: show up, be present, think critically about what I am saying, and ask questions. **You are expected to take your own notes.** If you happen to miss a class, get notes from someone in your study network. (However, I emphasize that you are personally responsible for whatever material you missed.) **I highly recommend taking notes by hand** as long as you are physically capable of doing so.

Recitation. Recitations are facilitated by the teaching assistant, and are typically either structured small-group work or brief lessons by the TA. They are typically (though not exclusively) oriented towards making progress on the current problem set.

Office hours. This is an opportunity for you and/or your group to come in and ask me in-depth technical questions about the course material and problem sets. You can book office hours with a link from the itsLearning webpage. If you see me in my office and my door is open, feel free to ask me if I have time to chat (I may say no, of course!).

My office is Ø17-601a-2, located at the end of the top floor of Knoldene Ø.

Deliverables and grading

Problem sets. There are 7 written problem sets assigned during the course of the semester; you will have about two weeks for each one. (Problem sets are assigned Mondays and due at midnight on Fridays at the end of the next week.) There will be problem sets due at the end of weeks 10, 11, 13, 16, 18, 20, and 22. (Note that week 14 classes are not in session.)

Problem sets will be distributed as physical copies, pdfs, and as LaTeX source code. You may work on them in whatever medium you choose; however, the final submission must be as a pdf. You are **encouraged but not required** to submit in groups. Each problem set will be graded on a 7-point scale, with a rubric

indicating what kind of solution earns what kind of score. (If you submit as a group, all members of the group get the same score.)

Use of generative AI, Stack Exchange, or any other way of receiving intelligent help online is **expressly forbidden**.

Final exam. There will be a written final exam on Monday, June 1. I am told that “written” means “on the computer,” which was not my intention when I wrote the course specifications. So I am trying to see what can be done; however, there will be a written exam.

Course grades. Your grade in the course will be determined by averaging your score on the problem sets (dropping the lowest two) with your score on the final. Problem sets and exams are weighted 75/25 respectively. This means each of your top five problem sets contributes 15% to your final grade and your final exam contributes 25%.

Tentative schedule

This comes with the usual disclaimer that it’s subject to change. Note there are no classes during Week 14.

Week	Topic	Book chapter	Problem sets
9	Introduction and discrete mathematics review		
10	Logic of “all p are q ”	1	Homework 1 due
11			Homework 2 due
12	... plus verbs and relative clauses	2	
13			Homework 3 due
15	Syllogistic logic	3	
16			Homework 4 due
17	Propositional logic	5	
18			Homework 5 due
18	First-order logic	(own notes)	
19			Homework 6 due
20			
21			Homework 7 due

The final exam is scheduled on Monday, June 1, in Week 22.